Practical no. 6A

Aim: a. Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

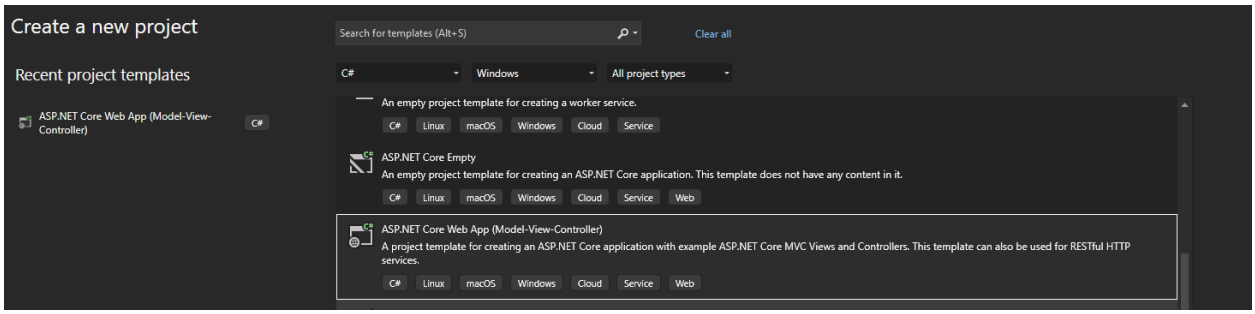
Software Required:

Visual Studio 2026

Procedure:

Step 1: Create New MVC Project

1. Open Visual Studio 2022.
2. Click Create a New Project.
3. Select ASP.NET Core Web App (Model-View-Controller).



4. Click Next.
5. Enter Project Name: Practical6b
6. Select .NET 8.0.
7. Click Create.

Step 2: Open Index.cshtml

Open:

Views → Home → Index.cshtml

Delete the existing code and enter:

```
@{
    ViewData["Title"] = "College Information Portal";
}

<div class="text-center">
    <h1>College Information Portal</h1>
    <p>Welcome to our College Information Portal</p>
</div>
```

```
<hr />

<h2>About College</h2>
<p>
   Our college provides quality education and develops
   technical and professional skills among students.
</p>

<h2>Courses Offered</h2>
<ul>
   <li>B.Sc. Information Technology</li>
   <li>B.Sc. Computer Science</li>
   <li>B.Com.</li>
   <li>B.A.</li>
</ul>

<h2>College Facilities</h2>
<ul>
   <li>Computer Laboratory</li>
   <li>Library</li>
   <li>Internet Facility</li>
   <li>Sports Facility</li>
</ul>

<h2>Contact Us</h2>
<p>
   Email: info@college.edu<br />
   Phone: 0250-1234567<br />
   Address: College Road, Maharashtra
</p>
```

Step 3: Open site.css

Open:

wwwroot → css → site.css

Add the following CSS:

```
body {
   font-family: Arial, sans-serif;
}

h1 {
   color: #0d6efd;
```

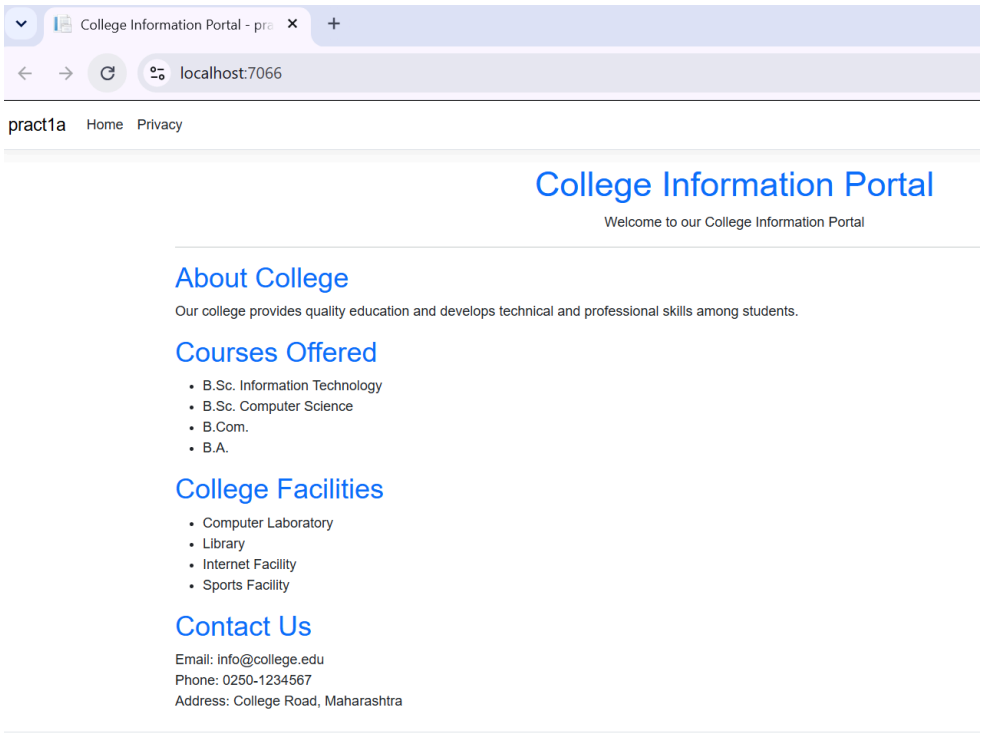
```
}

h2 {
    color: #0d6efd;
}

.text-center {
    text-align: center;
}
```

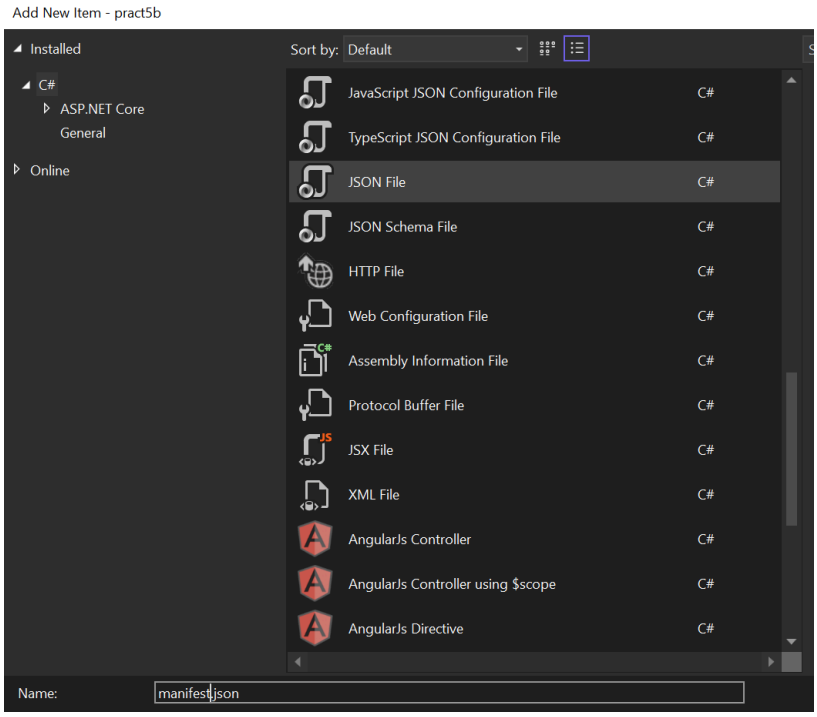
Step 4: Run and Test the Website

Press Ctrl + F5



Step 5: Create manifest.json

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select json file.
4. Enter File Name: **manifest.json**
5. Click Add.



Enter:

```
{
 "name": "College Information Portal",
 "short_name": "College Portal",
 "start_url": "/",
 "display": "standalone",
 "background_color": "#ffffff",
 "theme_color": "#0d6efd",
 "icons": [
   {
    "src": "/icons/icon-192.png",
    "sizes": "192x192",
    "type": "image/png"
   },
   {
    "src": "/icons/icon-512.png",
    "sizes": "512x512",
    "type": "image/png"
   }
 ]
}
```

Step 6: Add App Icon

1. Right Click → **wwwroot** folder

2. Click Add → New Folder
3. Enter Folder Name: **icons**
4. Paste one image in the **icons** folder.

Step 7: Edit _Layout.cshtml
Open:

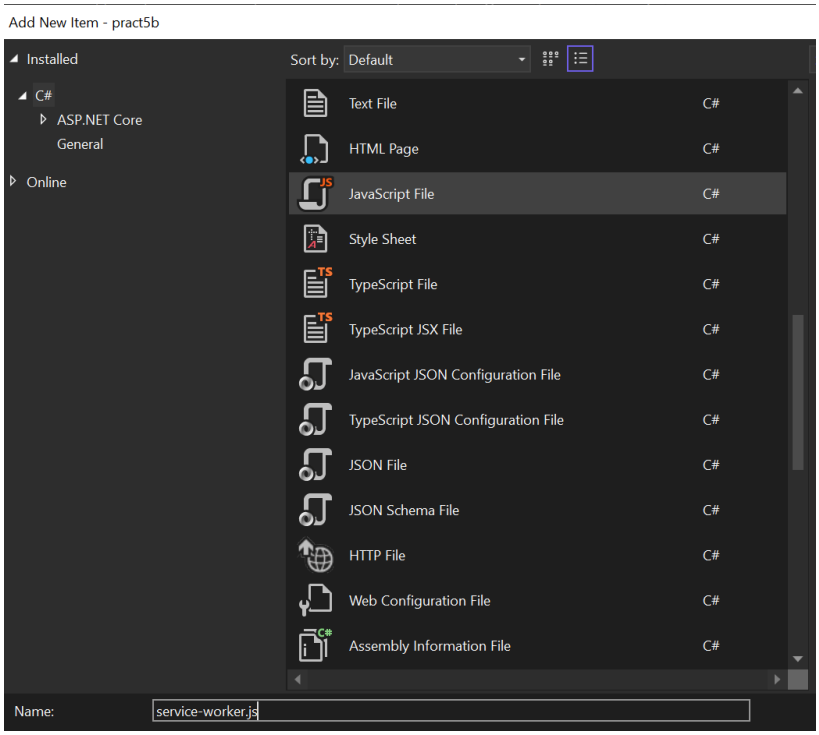
Views → Shared → _Layout.cshtml

Inside the <head> section, add:

```
<link rel="manifest" href="/manifest.json">
<meta name="theme-color" content="#0d6efd">
```

Step 8: Create Service Worker

1. Right Click → **wwwroot** folder
2. Click Add → New item
3. Select JavaScript file.
4. Enter File Name: **service-worker.js**
5. Click Add.



Write the following code in the service-worker.js file.

service-worker.js

```
const CACHE_NAME = "college-portal-v1";

const FILES_TO_CACHE = [
    "/",
    "/css/site.css",
    "/manifest.json"
];

self.addEventListener("install", function (event) {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(function (cache) {
                return cache.addAll(FILES_TO_CACHE);
            })
    );
});

self.addEventListener("fetch", function (event) {
    event.respondWith(
        caches.match(event.request)
            .then(function (response) {
                return response || fetch(event.request);
            })
    );
});
```

Step 9: Register the Service Worker
Open:

wwwroot → js → site.js

Add:

```
if ("serviceWorker" in navigator) {

    window.addEventListener("load", function () {

        navigator.serviceWorker.register(
            "/service-worker.js"
        );

    });

}
```

Step 10: Run the application
Press Ctrl + F5 and open the application in Chrome or Edge.

Step 11: Check the Manifest
Press F12 → Application → Manifest.

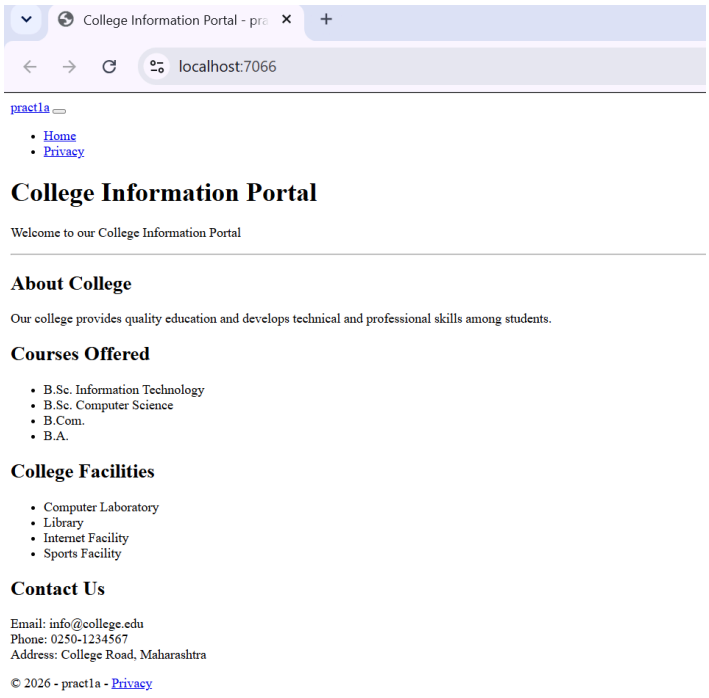
Check that the browser displays:

- College Information Portal
- College Portal
- Start URL: /
- Display: standalone
- Application icons

Step 12: Check the Service Worker
Go to F12 → Application → Service Workers.

Check that service-worker.js is registered and running.

Step 13: Test Offline Functionality
Open F12 → Network. → Select the Offline option and refresh the page.



Step 14: Install the PWA

1. Click Chrome Menu: ⋮ (Three Dots)
2. Select: Cast, Save and Share
3. Then click: Install Page as App or Install College Information Portal

How to Uninstall the PWA

1. Open installed application.
2. Click: ⋮ Menu
3. Select: Uninstall College Information Portal
4. Click: Remove